Exercises

Problem 4.1: Vanishing mean score

Let $p_\varphi(x)$ be a familiy of probability distributions over x parametrised by some real parameter φ. Define the *score* as $S_\varphi(x) = \frac{\partial}{\partial\varphi}\log(p_\varphi(x))$. Show that the mean value of the score over the distribution $p_\varphi(x)$ is zero.

Problem 4.2: Additivity of Fisher information

Prove that the Fisher information is additive for independent measurements. That is, when φ is inferred from two measurements X and Y with $p_\varphi(x,y) = p_\varphi(x)p_\varphi(y)$ show that $\mathcal{F}_\varphi(X,Y) = \mathcal{F}_\varphi(X) + \mathcal{F}_\varphi(Y)$.

Problem 4.3: MSE, variance, and bias

Prove the relation (4.12) between the MSE, variance, and bias of estimators.

Problem 4.4: NOON-state interferometry

Consider an N-photon NOON-state in two modes A and B.

$$|\psi_{NOON}\rangle_{\mathrm{AB}} = \frac{1}{\sqrt{2}}(|N,0\rangle_{\mathrm{AB}} + |0,N\rangle_{\mathrm{AB}}).$$

A phase shift φ is applied to mode B and the two modes are subsequently interfered on a balanced beam splitter. The unitary encoding the phase is $\hat{U}_\varphi = e^{-i\varphi\hat{n}_{\mathrm{B}}}$, where $\hat{n}_{\mathrm{B}}$ is the number operator on mode B. At each output of the beam splitter, the photon number is measured.

(a) Show that the state after the phase shift and before the beam splitter is

$$\frac{1}{\sqrt{2}}(|N,0\rangle_{\mathrm{AB}} + e^{-i\varphi N}|0,N\rangle_{\mathrm{AB}}).$$

(b) Show that in one of the output detectors, the probabilities for observing even and odd numbers of photons are $p_{\mathrm{even}} = \cos^2(\varphi N/2)$ and $p_{\mathrm{odd}} = \sin^2(\varphi N/2)$, respectively.

(c) Show that the Fisher information corresponding to this binary measurement of photon number parity equals $\mathcal{F}_\varphi = N^2$.

Problem 4.5: QFI for pure states

Consider a noiseless quantum parameter estimation scheme where the parameter is encoded unitarily. For a pure input state $|\psi\rangle$, the output state is

$$|\psi_\varphi\rangle = \hat{U}_\varphi|\psi\rangle = e^{-i\varphi\hat{H}}|\psi\rangle,$$

where $\hat{H}$ is some hermitian operator.

(a) Show that the QFI can be expressed in terms of the variance of $\hat{H}$ in the input state $|\psi\rangle$ as

$$\mathcal{Q}_\varphi = 4\left[\langle\psi|\hat{H}^2|\psi\rangle - \langle\psi|\hat{H}|\psi\rangle^2\right] = 4\langle\Delta^2\hat{H}\rangle_\psi.$$

(b) Assume that the system is finite dimensional and that the smallest and largest eigenvalues of $\hat{H}$ are $\lambda_{\min}$ and $\lambda_{\max}$, respectively. Show that the

optimal input state maximising the QFI is given by an equal superposition of the corresponding eigenstates, i.e.

$$\underset{|\psi\rangle}{\operatorname{argmax}}\, \mathcal{Q}_\varphi = \frac{1}{\sqrt{2}}(|\lambda_{\min}\rangle + |\lambda_{\max}\rangle).$$

Is the relative phase between $|\lambda_{\min}\rangle$ and $|\lambda_{\max}\rangle$ important?

(c) Show that the value of the QFI for the optimal input state is $(\lambda_{\max} - \lambda_{\min})^2$.

Problem 4.6: SQL and Heisenberg scaling for pure states

Consider a noiseless quantum parameter estimation scheme with unitary encoding, as in Prob. 4.5, and consider that there are N identical, non-interacting copies of the system.

(a) Argue that the encoding unitary for the total system can be written $\hat{U}_\varphi^{(N)} = e^{-i\varphi \hat{H}^{(N)}}$ with

$$\hat{H}^{(N)} = \hat{H}_1 + \hat{H}_2 + \cdots \hat{H}_N,$$

where $\hat{H}_k$ is $\hat{H}$ on the k'th system and identity on all the others, that is $\hat{H}_1 = \hat{H} \otimes \mathbb{1} \otimes \cdots \otimes \mathbb{1}$ etc.

(b) Show that if the N-partite system is initially prepared in a product of identical states $|\psi^{(N)}\rangle = |\psi, \ldots, \psi\rangle = |\psi\rangle \otimes \cdots \otimes |\psi\rangle$ then the QFI for the total system is given by

$$\mathcal{Q}_\varphi^{(N)} = N\mathcal{Q}_\varphi.$$

Argue that this implies a variance in the best estimate of φ scaling as the standard quantum limit (SQL)

$$\langle \Delta^2 \tilde{\varphi} \rangle \geq \frac{1}{\nu N (\lambda_{\max} - \lambda_{\min})^2},$$

where ν is the number of independent repetitions of the whole experiment.

(c) Show that that the smallest and largest eigenvalues of $\hat{H}^{(N)}$ are $N\lambda_{\min}$ and $N\lambda_{\max}$, respectively.

(d) Show that the optimal (entangled) input state for the N-partite system is

$$\frac{1}{\sqrt{2}}(|\lambda_{\min}, \ldots, \lambda_{\min}\rangle + |\lambda_{\max}, \ldots, \lambda_{\max}\rangle.$$

Argue that the corresponding variance in the best estimate of φ scales as the Heisenberg limit

$$\langle \Delta^2 \tilde{\varphi} \rangle \geq \frac{1}{\nu N^2 (\lambda_{\max} - \lambda_{\min})^2}.$$